

Practical 8

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

```
GNU nano 8.7          lru.c
#include<stdio.h>

int main()
{
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3], time[3];
    int n = 12;
    int i,j,pos;
    int hits=0,faults=0;

    for(i=0;i<3;i++)
        frames[i]=-1;

    for(i=0;i<n;i++)
    {
        int flag=0;
        for(j=0;j<3;j++)
        {
            if(frames[j]==pages[i])
            {
                hits++;
                time[j]=i;
                flag=1;
                break;
            }
        }
        if(flag==0)
        {
            int lru=time[0];
            pos=0;
            for(j=1;j<3;j++)
            {
                if(time[j]<lru)
                {
                    lru=time[j];
                    pos=j;
                }
            }
            frames[pos]=pages[i];
            time[pos]=i;
        }
    }

    printf("Hits: %d\n", hits);
    printf("Faults: %d\n", faults);
    printf("Percentage Hit: %.2f\n", (float)hits/n*100);
}
```

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^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_

```

        break;
    }
}

if(flag==0)
{
    int lru=time[0];
    pos=0;

    for(j=1;j<3;j++)
    {
        if(time[j]<lru)
        {
            lru=time[j];
            pos=j;
        }
    }

    frames[pos]=pages[i];
    time[pos]=i;
    faults++;
}

printf("Page Hits = %d\n",hits);
printf("Page Faults = %d\n",faults);

printf("Hit Percentage = %.2f%%\n",(hits*100.0)/n);
printf("Fault Percentage = %.2f%%\n",(faults*100.0)/n);

return 0;
}

```

^G Help ^O Write Out ^F Where Is ^K Cut
 ^X Exit ^R Read File ^_ Replace ^U Paste

```

Dell@aryaaa MSYS ~
$ nano lru.c

Dell@aryaaa MSYS ~
$ gcc lru.c -o lru

Dell@aryaaa MSYS ~
$ ./lru
Page Hits = 2
Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%

Dell@aryaaa MSYS ~
$ |

```